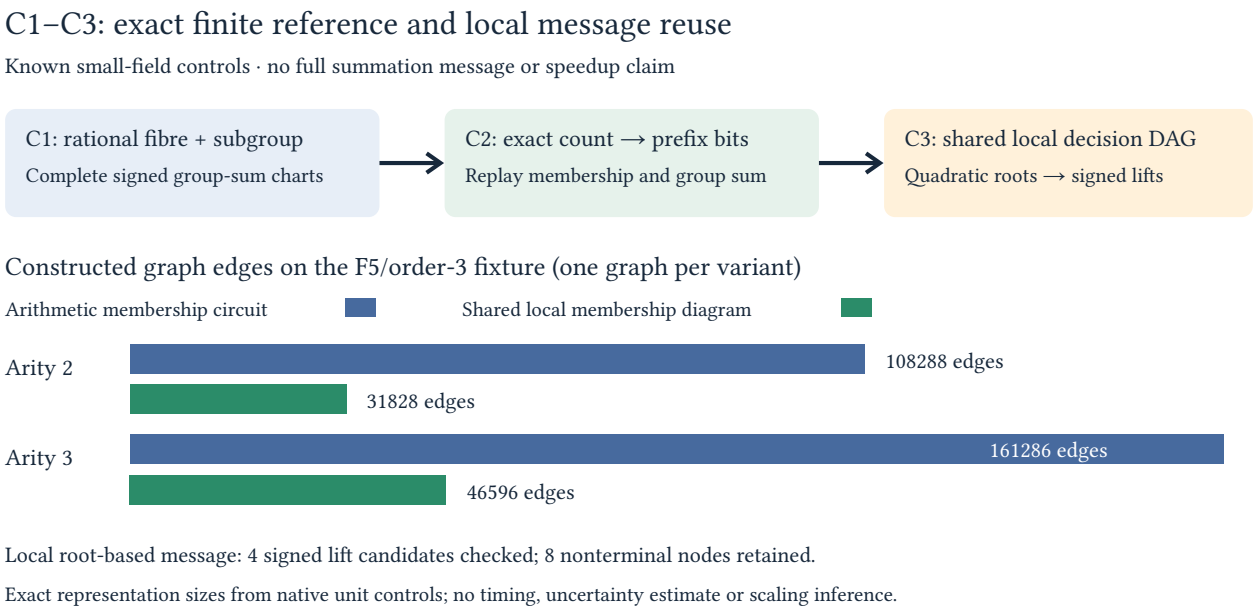
Structured prime-field IC: C1–C3 implementation controls

2026-10-30 · bounded native references; general goals remain partial

C1 — Exact predicate. A rational two-isogeny fibre constraint, canonical rational coordinates and actual subgroup membership are combined with complete point-summation charts. The special family requires rational two-torsion; it is not a construction for arbitrary prime-order curves or all PKM auxiliary maps.

C2 — Count-guided recovery. Exact counter queries select interface bits, then native arithmetic rechecks every signed summand and the target sum. Repetitions, inverse pairs and intermediate identity sums are retained. Resource exhaustion means unknown. The functional leaf still enumerates inputs, so this is not a new grid-free complexity bound.

C3 — Local message reuse. A shared reduced decision diagram represents local membership. One constructor exhausts coordinate encodings; the other solves quadratic fibres, lifts both rational signs and checks subgroup membership. The root constructor avoids the field scan. Setup and storage are charged. It is not a compact contracted message for the full summation predicate.



Checked scope. Exact counts and witnesses agree with direct addition on all three targets of the order-3 subgroup of $y^2=x^3+1$ over F5, at arities2 and3. Two-point counts are2,1,1; three-point counts2,3,3. On $y^2=x^3+x$ over F19, order5, all signed base-pair/target predicates agree. Every single image fibre in both fields compares the root-based message, exhaustive message and direct membership, including poles, empty fibres and noncanonical coordinates.

Limits and evidence. The successful query controls use fresh500million-step envelopes; the default100million is unchanged. Earlier compile errors and resource failures remain in checks/. Graph edge counts above are structural software diagnostics, not benchmark or cold-total costs. No unknown-target solve, isolated timing panel, independent-host replay, Lean proof or new mathematical result is asserted. Source hashes, raw checks, exact parameters, reproduction commands and the open full-message obligation are in README.md.

Sources. PKM2016 sections3.1–3.4; Hexagon2610.00594v1,2610.00600v1 and 2610.00613v1; Kosters–Yeo1503.08001v3; existing prime-IC, isogeny-fibre and torus studies. Cached primary texts were used; fresh web requests failed, and the earlier2018 full-text gap remains. Global novelty and an IC improvement are unestablished. Existing leaderboard/frontier graphs gain no qualified data.